English to Moroccan Darija (in Arabizi) Machine Translation via a Transformer Architecture

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***Abstract*—Moroccan Darija is a language descended from Arabic characterized by the absence of standardized grammar and orthography, making its machine translation particularly complex. This work presents the development of an English–Darija Transformer machine translation model, optimized under a strict constraint of 100 million parameters to enable offline execution on smartphones. We leverage a dataset of 280k sentence pairs generated through knowledge distillation from Gemini 2.5 Flash. To mitigate the high lexical variability of Darija, we implement a single, joint English-Darija tokenizer with a shared vocabulary (32k) coupled with BPE Dropout regularization. The architecture integrates Weight Tying and Pre-Layer Normalization, reaching a total of 98M parameters. Given the absence of standard BLEU benchmarks adapted to non-standardized languages, evaluation is conducted via an LLM-as-a-judge approach, demonstrating remarkable performance against the structural challenges of the language.**

1. Introduction

Machine translation for low-resource or non-standardized languages represents a major challenge in Natural Language Processing (NLP). Moroccan Darija (Moroccan Arabic) fits perfectly into this problem: it is an essentially oral language, characterized by a lack of strict grammatical and spelling norms. While usually written in the Arabic alphabet, we focus here on direct translation into Arabizi (Latin characters), as a simple transliteration approach fails (e.g., transliteration maps the word ”car” into ”twmwbl” rather than ”tomobil”). Most online resources in this language and alphabet stem from digital communications, which has allowed large language models to assimilate it effectively.

The objective of this project is to design a high-performing English-to-Darija translation system, while respecting a strong hardware constraint: the final model must not exceed the threshold of 100 million parameters to guarantee its viability and smooth execution directly on smartphones.

2. Dataset Construction and Normalization

One of the primary bottlenecks in developing models for Darija is the unavailability of parallel corpora of sufficient size, save for the DODa project, which contains few samples written in Arabizi.

2.1. Generation via Knowledge Distillation

To overcome this lack of resources, I implemented a knowledge distillation strategy from the state-of-the-art `Gemini 2.5 Flash` model, which writes in this language with high accuracy. This method yielded a high-quality parallel corpus comprising approximately 280,000 English–Darija sentence pairs for an optimized cost of around 70 euros. Data samples are provided in the appendix. The dataset was split using a standard partition: 95% for training and 5% for validation. The corpus remains modest in size, as machine translation datasets typically span millions of sentences.

2.2. Preprocessing and Sequence Management

Text normalization focused on systematic lowercasing across the entire corpus. Given the morphosyntactic structure of the generated sentences, the maximum sequence length was set to 256 tokens, which is generally sufficient to translate daily conversational sentences.

3. Tokenization Strategy

The orthographic variability of Darija constitutes a major hurdle for classical tokenization algorithms (e.g., *bezzaf*, *bezaf*, or *bzaf* all meaning ”a lot”). Throughout the project, the tokenizer proved to be the primary bottleneck to the model’s progression.

3.1. Limitations of the Disjoint Approach

A first attempt relied on using two distinct Byte Pair Encoding (BPE) tokenizers: an English sub-corpus with a vocabulary of 32k tokens, and a Darija sub-corpus with a vocabulary of 48k tokens. This method showed its limits by treating spelling variations as entirely independent entities, thereby fragmenting the semantic space. To illustrate, the model might hesitate among the three spellings *bezzaf*, *bezaf*, or *bzaf*, diluting their probability mass and ultimately selecting an inadequate token.

3.2. Optimization via Shared Vocabulary and BPE Dropout

To resolve this issue, I opted for a single, shared tokenizer combining English and Darija with a restricted vocabulary size of 32k tokens. This unified approach promotes better sub-word decomposition:

$$\text{b—ez—zaf} \approx \text{b—e—zaf} \approx \text{b—zaf} \quad (1)$$

Furthermore, introducing the *BPE Dropout* technique during training forces the model to encounter multiple possible segmentations for the same word (for instance, alternating between `b|ez|zaf` and `bez|zaf`), reinforcing model robustness against spelling variations. An example of corpus segmentation is illustrated below:

```
kanbghi| nt|m|acha| f| zzna|9|i| dyal|
                    l|7|ay|
```

4. Model Architecture and Training

The model is based on the standard Transformer Encoder-Decoder architecture but integrates critical structural modifications to improve convergence and efficiency.

4.1. Architectural Modifications

Weight Tying: The weight matrices of the input and output embeddings are shared. This not only reduces the overall parameter count but also improves training stability. Indeed, without weight tying, the output linear layer tends to heavily project toward common tokens (such as punctuation or stop words) during the first training iterations, causing a representation collapse that takes a long time to recover from.

Pre-Layer Normalization (Pre-LN): In contrast to the original Post-LN architecture, normalization is applied before each sub-layer (attention and feed-forward) as well as at the output of the encoder and decoder, rather than on the residual path. This stabilizes the gradients.

4.2. Hyperparameters

To maximize the utility of the imposed size limit, the model was configured with the hyperparameters presented in Table 1.

TABLE 1. ARCHITECTURAL HYPERPARAMETERS OF THE SELECTED MODEL

Hyperparameter	Value / Configuration
Encoder Layers	10
Decoder Layers	12
Embedding Dimension (d_{model})	512
Attention Heads per Layer (n_{head})	8
Max Sequence Length	256 tokens
Total Parameters	**98 Million**

The respective depths of the encoder and decoder allow for better representation learning on a language as noisy in its spelling as Darija. A high dropout rate (0.2) was also applied within the residual layers and attention matrices to counteract the small dataset size.

4.3. Training Configuration

The loss function used is *Cross-Entropy Loss* configured with a *Label Smoothing* coefficient of 0.1 to prevent model overconfidence regarding different spellings of the same word. Optimization utilizes the `AdamW` algorithm combined with a *Weight Decay* coefficient of 0.01. Gradients are bounded via *Gradient Clipping* at a maximum norm of 1.0 to prevent gradient explosions. A *Learning Rate Scheduler* featuring a linear warmup followed by a Cosine Annealing decay guided the model to optimal convergence.

5. Inference

During inference, token selection leverages a frequency penalty (*Frequency Penalty = 0.6*) to prevent repetitive patterns, combined with an exploration of classic decoding strategies: *Top-K sampling*, *Top-P (Nucleus) sampling*, and *Beam Search* to maximize the global coherence of the generated sentences. Beam Search decoding coupled with the Frequency Penalty was retained for final evaluations.

6. Evaluation and Perspectives

The lack of standardized automated evaluation benchmarks (such as validated BLEU tests) for Darija constitutes an inherent limitation for non-standardized languages. To bypass this limitation, the evaluation was conducted by adopting the modern *LLM-as-a-judge* methodology, where a large language model qualitatively assesses and ensures the semantic fidelity of the translations produced by our 98M-parameter model. In this case, Gemini 3.5 Pro was used to validate prediction coherence. Preliminary results are highly promising, showing excellent capture of local idioms.

Future work will focus on handling proper nouns, which are often out-of-vocabulary words unseen in the dataset, posing a challenge for the model.

This work can be generalized to other languages with non-standardized orthography and grammar, and low online resources. For some of these languages, models such as Meta's *NLLB* or Google's *T5* show remarkable baseline results.

7. Appendices

7.1. Observation of Trained Embeddings

To validate the semantic alignment achieved by the model during training, a two-dimensional Principal Component Analysis (PCA) was applied to the vector embedding space.

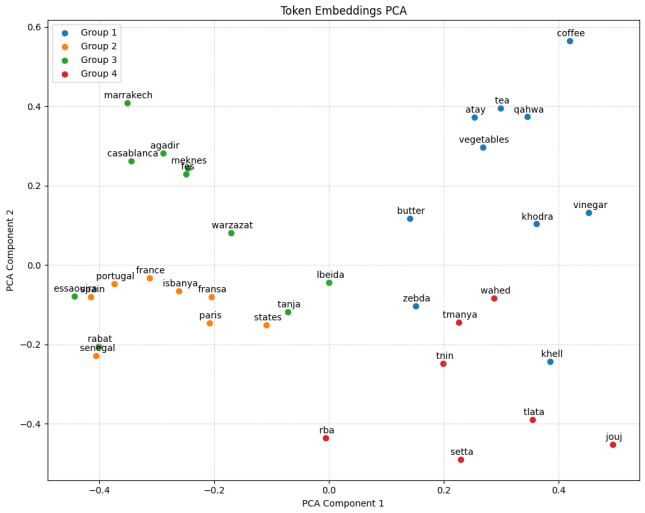


Figure 1. PCA visualization of thematic clusters within the embedding space.

Figure 1 highlights the natural formation of semantic clusters grouping concepts by similar themes, demonstrating a logical organization of the latent space.

Furthermore, cross-lingual alignment was geometrically verified. As illustrated in Figure 2, the projections reveal that shared named entities (such as country names) cluster in close proximity across both languages, confirming the efficiency of the single tokenizer and the shared vocabulary strategy.

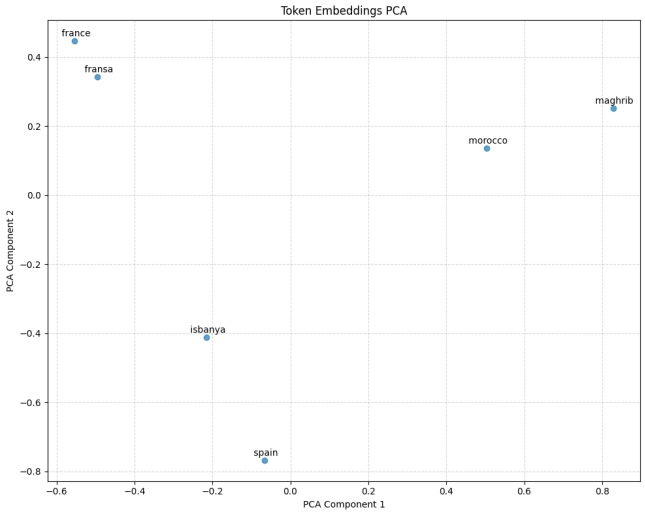


Figure 2. Geometric alignment of identical semantic concepts (countries) across both languages.

7.2. Dataset Extracts

The dataset constructed via distillation includes sentence pairs of varying lengths. Table 2 presents examples of short and long extracts from the English–Darija parallel corpus.

7.3. Generated Translation Examples

The generalization capabilities of the 98M-parameter model are evaluated on complex utterances with a heavy cultural or contextual load (Table 3).

TABLE 2. EXAMPLES OF SHORT AND LONG SENTENCE PAIRS FROM THE TRAINING DATASET

English (Source)	**Moroccan Darija (Target)**
Short sentence examples	
I'm not familiar with the route to Rabat, so I'll rely on your experience. Could you ensure we take the quickest and safest way?	Ma 3arfch triq l Rabat, dakchi 3lach ghan3tamed 3la khibrtek. Momkin tdamen liya nakhdo asra3 w a2man triq?
My budget is limited, I can't pay more than 55 dirhams.	Lbudget diali m7doud, ma nqderch nkhallas kter men 55 dirham.
Long sentence example	
The international community's increasing recognition of the Sahara's intrinsic link to Morocco's historical and cultural identity, alongside its strategic importance for regional security and stability, means that any resolution must fully integrate the specificities of this unique region within the broader framework of Moroccan sovereignty, ensuring that the local populations are empowered to manage their affairs while remaining an integral part of the Kingdom, thus securing a peaceful and prosperous future.	L2i3tiraf lli kaykber mn lmujtama3 ddawli b rraba6 l2asassi dyal ssahra b lhowiya ttariikhiya w ttakafiya dyal lmghrib, m3a l2ahmiya listratijiya dyalha l2amn w listi9rar ljiihawi, kay3nii bali ay 7all khassu yddmaj bchakl kamil lkhousousiyat dyal had lmin6a9a lfaaridda dakhil l2i6ar l3am dyal ssiyada lmaghribiya, kayaddamman tamkin nnas lma7alliyin bach yssayro chomounhom m3a lba9a2 kjjuz2 la yatajaza2 mn lmamlakka, w bttali kayaddamman musta9bal ssalamii w raakhii.

TABLE 3. EXAMPLES OF TRANSLATION OUTPUTS OBTAINED AT INFERENCE

Source	**Model Translation**
Can you show me the way to the old medina? I'm completely lost and my phone's battery is dead.	wach t9der twerini tri9 l lmedina l9dima? ana telft kaml w tele dyali ma b9ach fih la batterie.
I spent nearly forty-five minutes this morning arguing with the merchant at the weekly souk over the price of fresh mint and orange blossom water needed for the family gathering, entirely forgetting that my grandmother expected me back hours ago to help roll the semolina grains for the Friday couscous, which meant I had to hurriedly catch a shared grand taxi under the blazing midday heat just to avoid her legendary disapproval.	dwezzt chi khamsa w arba3in dqiqa had sba7 wana kan3awd m3a lbiya3 f souq l'ousbou3i 3la taman dyal na3na3 jdid w ma zhar li khasso l jam3a dyal la famille, nssit ga3 belli jaddati kant katssennani men qbel swaye3 bach n3awen ndawwer 7boub smida l couscous dyal nhar ljomu3, dakchi li khasni khassni nchadd taxi kbir mouchtarak ta7t skhouna dyal nos nhar ghir bach netfadda rafdha l'os6ori.